HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python** scripts to automate data extraction and transformation for manufacturing reports, reducing manual processing time by approximately **40%**.
- Assisted in the troubleshooting and optimization of **PLC-controlled** equipment, identifying and resolving communication bottlenecks that improved operational uptime.
- Supported the integration and commissioning of **robotics and automated manufacturing equipment**, contributing to successful New Product Introduction (NPI) phases.
- Maintained and enhanced **Java Spring Boot** applications supporting internal manufacturing tools, implementing bug fixes and minor feature enhancements.
- Wrote **SQL** queries for data validation and ad-hoc reporting against production databases, ensuring data integrity for operational dashboards.
- Containerised **Java Spring Boot** applications using **Docker** for consistent deployment across development and testing environments.
- Participated in **Agile/Scrum** ceremonies, including sprint planning and daily stand-ups, to coordinate development efforts and track progress.

SKILLS

Programming Languages – Python, Java, SQL, C#

Automation & Control – PLC Programming, Siemens, Allen-Bradley, Robotics, Industrial Automation

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Development Tools & Practices – Git, Jenkins, Jira, Confluence, Agile, Scrum, Code Reviews

Databases – PostgreSQL, MySQL, Oracle

Protocols – Ethernet/IP, Modbus, Profinet, OPC-UA

PROJECTS

Manufacturing Data Automation Suite

- Developed a suite of **Python** scripts to automate the collection and aggregation of manufacturing data from disparate sources, including sensor logs and production databases.
- Implemented data validation checks using **SQL** to ensure the accuracy and completeness of extracted data before loading into reporting tables.
- Created automated reports summarizing key performance indicators, reducing the time required for weekly operational reviews by **~35%**.

Automated System Monitoring and Alerting

- Built a **Java Spring Boot** application to monitor the status of critical automated manufacturing systems, integrating with **AWS CloudWatch** for metric collection.
- Configured alerting mechanisms to notify operations teams of system failures or performance degradation, ensuring prompt response to production issues.
- Designed a simple REST API for querying system status, facilitating integration with broader manufacturing execution systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)